Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (EC) Examination

March - April 2018

EC 305.02 / 305.01 Microcontroller & Application

Date: 05/04/2018, Thursday **Time: 01:30 p.m. To 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q-1 (a) Differentiate low-level language and high-level language. **[03]**

(b) Explain function of following 8051 pins. **[04]**

1. ALE 2. EA 3. PSEN 4. RST

Q-2 **Answer the following Questions. (Any Two)** **[14]**

(a) Interface 16x2 LCD with 8051. Write an 8051 C program to display "MARCH" in first line and "2018" in second line.

(b) Explain memory organization of 8051 microcontroller.

(c) Write an 8051 program to transfer serially the message "BEST OF LUCK" continuously at a 9600 baud rate.

Q-3 **Answer the following Questions. (Any Two)** **[14]**

(a) Write an 8051 C program to toggle bits of P1 with 50 ms delay.

(b) Write an 8051 program to generate square wave of 2500Hz on P2.7. Use Timer 1, mode 2 to create the delay. Assume XTAL= 11.0592MHz.

(c) A switch is connected to pin P1.2. write an 8051 program to monitor switch and create the following frequency on pin P1.7:

If Switch = 0: 500 Hz

Switch = 1: 750 Hz, use timer 0 mode 1.

SECTION – II

Q-4 **Do as Directed.** **[07]**

(a) How port 0 is different from the other ports of 8051 microcontroller? **01**

(b) Upon reset, the status of port 1 is ______________. **01**

(c) Assume that bit P2.3 is an input and represents the condition of an oven. If it goes high, it means that the oven is hot. Write an 8051 program to monitor the bit continuously. Whenever it goes high, send a high to low pulse to port P1.5 to turn on a buzzer. **05**

Q-5 Answer the following Questions. (Any Two) [14]

(a) Explain serial communication mode 0 and mode 1 in 8051. Also write the format of SCON register.

(b) Interface DC motor and push button switch (SW0) with 8051 microcontroller. Write an 8051 C program to rotate DC motor clockwise if SW=0 else anti clockwise.

(c) Interface 4x3 keypad with 8051. Write an 8051 program to detect pressed key.

Q-6 Answer the following Questions. (Any Two) [14]

(a) List Special Function Register (SFR) available with 8051. Draw and explain format of TMOD register.

(b) Write a program to transmit message "CHARUSAT UNIVERSITY" serially at baud rate of 4800, 8-bit data and 1 stop bit continuously.

(c) List merits and demirts of parallel and serial communication.

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